

Electricity Demand Forecasting in France

A weather- and calendar-driven multi-output forecasting pipeline

Marcos Lahoz

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Abstract

We forecast half-hourly electricity demand for France, its 12 administrative regions and 12 métropoles — 25 series in total — for the year 2022, using only weather observations and calendar information. We compare three models of increasing complexity (a seasonal climatology baseline, a multi-output neural network and per-series gradient-boosted trees) under a strict temporal validation protocol and the official challenge metric, the sum of the per-series RMSE. Gradient boosting is the strongest model, reducing the validation score by 39% relative to the seasonal baseline.

1 Introduction

Short-term electricity demand forecasting is central to grid operation and energy trading. This work addresses a data-challenge formulation: given historical demand (2017–2021) and meteorological data, predict the demand of 2022 for 25 geographic series simultaneously. Models are scored with

$$\text{score} = \sum_{s=1}^{25} \text{RMSE}_s,$$

the sum of the per-series root-mean-squared errors (lower is better). The same quantity is used as the training loss of the neural network, so the model optimises exactly the metric it is evaluated on.

2 Data

Three sources are provided: half-hourly demand for the 25 series (2017–2021); the 2022 timestamps to predict; and three-hourly observations from 40 Météo-France SYNOP stations distributed across the 12 regions (2017–2022). The 12 métropole series only begin partway through the training period, so the targets contain missing values; every stage of the pipeline is therefore made NaN-aware. Each target is mapped to the weather of its region (the national series uses the average across regions); stations are averaged per region and the resulting signal is resampled from three-hourly to the 30-minute grid of the demand series.

3 Methodology

3.1 Feature engineering

All features are available for the 2022 horizon (no leakage from future targets). *Calendar* features are computed in French local time: cyclical encodings of time-of-day, day-of-week and day-of-year, weekend and public holiday flags, and a yearly trend. *Weather* features include per-region temperature, humidity, wind and pressure, heating- and cooling-degree quantities (HDD/CDD), a national aggregate, and 24 h/48 h rolling temperatures that capture the thermal inertia of demand.

3.2 Validation protocol

Because the data is a time series, we use a strictly chronological split: models are trained on 2017–2020 and validated on the held-out year 2021. A random split would leak future information and over-estimate performance.

3.3 Models

- **Seasonal baseline** — the historical mean demand for each (month, weekday, half-hour) cell; a strong, weather-free reference.
- **Neural network (MLP)** — a two-hidden-layer perceptron with batch normalisation, dropout and Xavier initialisation, trained with mini-batches, early stopping, and the sum-of-RMSE loss (the challenge metric). The network predicts in standardised output space for stable optimisation while the loss is evaluated on the raw scale.
- **Gradient boosting** — one histogram-based gradient-boosting regressor per series, each trained on the rows where its target is observed.

4 Results

Table 1 and Figure 1 report the 2021 hold-out performance. Both learned models beat the seasonal baseline, and gradient boosting is clearly the best, lowering the challenge score from 11541 to 7037 (a 39% reduction) and roughly halving the mean absolute error. The neural network improves on the baseline but trails the trees, which handle the non-linear temperature–demand relationship and the heterogeneous scales of the 25 series particularly well.

Model	Challenge score	MAE
Seasonal baseline	11541.1	332.5
MLP (PyTorch)	9834.9	321.0
Gradient boosting	7036.6	211.1

Table 1: Performance on the 2021 hold-out year (sum of per-series RMSE; lower is better).

5 Conclusion

A compact pipeline built on temperature, calendar and degree-day features predicts the 25 demand series well across France. Under an honest temporal validation, per-series gradient boosting is the most accurate model and is used to generate the 2022 submission. Natural extensions include population-weighted national aggregation, lagged-demand features for shorter horizons, and a boosting/neural ensemble.

Reproducibility

The full code, models and instructions are available in the project repository. Results reproduce with `python scripts/evaluate.py -raw-dir data/raw -models baseline mlp gbm`.

References

- [1] Météo-France. *SYNOP observation data* (Données SYNOP essentielles OMM).

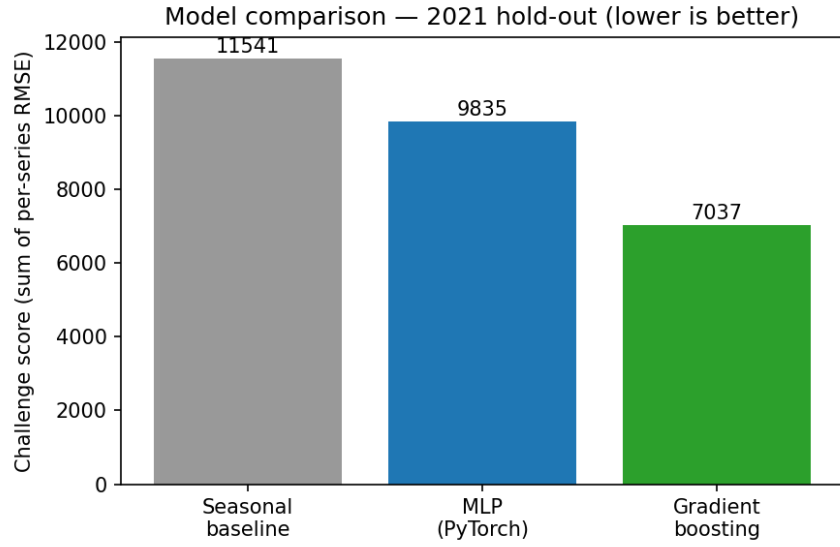


Figure 1: Validation challenge score by model on the 2021 hold-out.

- [2] Pedregosa et al. *Scikit-learn: Machine Learning in Python*. JMLR, 2011 (HistGradientBoostingRegressor).
- [3] Paszke et al. *PyTorch: An Imperative Style, High-Performance Deep Learning Library*. NeurIPS, 2019.